

Eigenfaces for Face Recognition

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Introduction

This assignment explores the use of Eigenfaces, a Principal Component Analysis (PCA) based approach for face recognition and dimensionality reduction of facial images. We follow the method outlined by Turk and Pentland, where the eigenfaces are derived from the eigenvectors of the covariance matrix of the training set of images.

Methodology

The Dataset

The Olivetti faces dataset was used for this implementation. The dataset consists of face images where:

- Each image is 64×64 pixels in grayscale
- Each image is flattened into a 4096-dimensional vector
- A set of M such images forms a data matrix X of size $M \times N$ (with $N = 4096$)

Computing the Face Space (Turk and Pentland Method)

Instead of directly performing an SVD on the large data matrix, the Turk and Pentland method exploits the fact that the number of images M is often much smaller than the dimensionality N of each image. The steps are as follows:

1. Compute the mean face:

$$\mu = \frac{1}{M} \sum_{i=1}^M x_i$$

where x_i is the i -th image as a vector of length N .

2. Subtract the mean face from each image to center the dataset:

$$X' = X - \mu$$

Here, X' is still $M \times N$.

3. Construct the covariance matrix in the reduced space:

$$L = X'X'^T \quad (\text{an } M \times M \text{ matrix})$$

4. Compute the eigenvalues and eigenvectors of L :

$$Lv_i = \lambda_i v_i$$

5. Since L is $M \times M$, we only obtain M eigenvectors. Each eigenvector v_i in this M -dimensional space can be mapped back to the original N -dimensional face space:

$$u_i = X'^T v_i$$

The vectors u_i are the eigenfaces.

6. Normalize each u_i so that it has unit length. The eigenfaces u_i correspond to the directions of maximum variance in the face space.

Dimensionality Reduction

After computing the eigenfaces, we can reduce the dimensionality:

- Select the top N eigenfaces corresponding to the largest eigenvalues.
- Project images onto this reduced eigenface subspace:

$$W = U_N^T (X' - \mu)$$

where U_N is the matrix of top eigenfaces.

- Each image is now represented by a vector of N dimensions instead of 4096.

In our implementation, we chose $N = 128$.

Image Reconstruction

To reconstruct an image from its lower-dimensional representation:

- Let w be the weight vector corresponding to the projection of an image x :

$$w = U_N^T (x - \mu)$$

- Reconstruct the image using:

$$\hat{x} = U_N w + \mu$$

Results

Top Eigenfaces

The following figure shows the top 10 eigenfaces computed from the dataset:

Image Reconstruction

Below is an example of an original image and its reconstruction using 128 eigenface components:

The reconstruction is of good quality while significantly reducing the dimensionality. This demonstrates the effectiveness of the Turk and Pentland eigenface method in capturing the essential features of face images and enabling efficient face recognition and analysis.

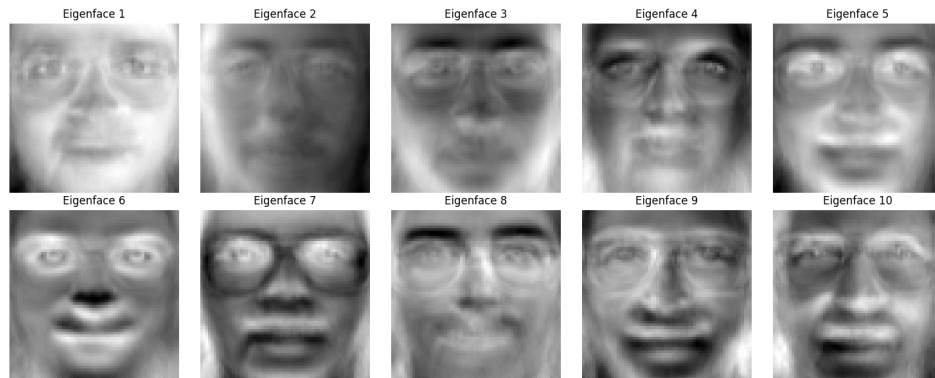


Figure 1: Top 10 eigenfaces computed from the Olivetti faces dataset

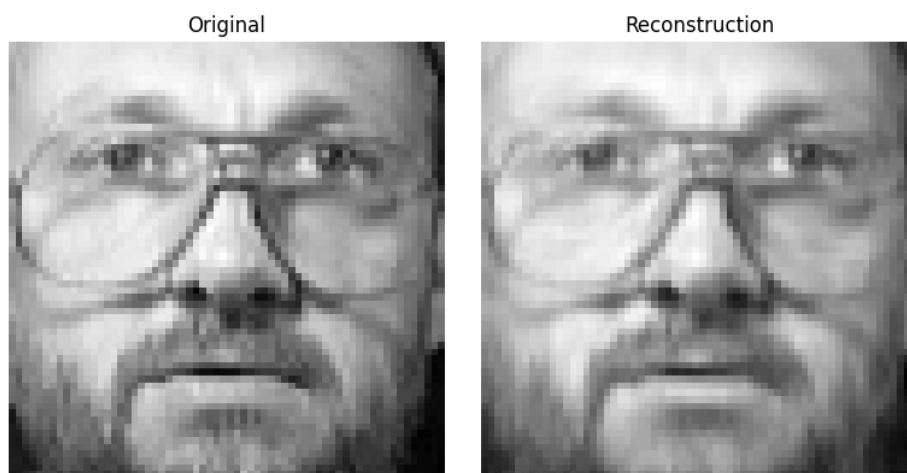


Figure 2: Original image (left) and its reconstruction (right)